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Weather Tracker API

REST API for tracking weather forecasts and observations for 38 cities. Built with Flask and MongoDB. Base URL: `http://127.0.0.1:5001/api/v1`. Write operations need a Bearer token from `/auth/login`. Admin-only routes return 403 for normal users. Test accounts alice, bob and carol use the password `password123` (test data only).

Auth

Endpoints for registering new users and logging in to obtain a JWT token.

POST Login

{{base_url}}/auth/login

StartFragmentLogs in and returns a JWT token valid for 2 hours EndFragment.

AUTHORIZATION Bearer Token

Token	{{json_web_token_07ks}}

Body raw (json)

```json
{"username": "alice", "password": "password123"}
```

POST Register alice

{{base_url}}/auth/register

Registers a new user account for alice and stores the returned JWT token.

Body raw (json)

json

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POST Register bad password

{{base_url}}/auth/register

Attempts to register with a password that fails validation rules. Expects 400 Bad Request.

Body raw (json)

json


```
{"username":"zed","email":"z@test.com","password":"123"}
```


Locations

Endpoints for listing, filtering, and retrieving weather-tracked city locations.

GET List locations

{{base_url}}/locations?pn=1&ps=5

Returns a paginated list of all tracked city locations.

PARAMS

pn	1
ps	5

GET Filter by country

{{base_url}}/locations?country=UK

Returns all tracked locations filtered by the given country code.

PARAMS

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GET Nearby

{{base_url}}/locations/nearby?lat=54.6&lng=-5.9&radius_km=500

Returns locations within a given radius (km) of the specified latitude and longitude.

PARAMS

lat	54.6
lng	-5.9
radius_km	500

GET Get one

{{base_url}}/locations/{{city_id}}

Returns the details of a single location by its ID ({{city_id}}).

GET Invalid id

{{base_url}}/locations/abc

Requests a location using a non-numeric ID. Expects a 400 or 404 error response.

Forecasts

Endpoints for creating, retrieving, and deleting weather forecasts for a location.

POST Add forecast

{{base_url}}/locations/{{city_id}}/forecasts

{{base_url}}/locations/{{city_id}}/forecasts

Creates a new forecast for the tracked city. Saves the new forecast ID to .

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Token

Body raw (json)

```
json

{"date": "", "temp_max": 12, "temp_min": 5, "humidity": 80, "wind_kph": 20, "precip_mm":
```

POST Duplicate forecast

{{base_url}}/locations/{{city_id}}/forecasts

Attempts to add a forecast for a date that already exists. Expects 409 Conflict.

Body raw (json)

```
json

{"date":"","temp_max":12,"temp_min":5,"humidity":80,"wind_kph":20,"precip_mm":3,"condition"
```

POST Bad humidity

{{base_url}}/locations/{{city_id}}/forecasts

Submits a forecast with an invalid humidity value (500). Expects 400 Bad Request with the humidity field named in the error details.

Body raw (json)

```
json

{"date":"2030-01-02","temp_max":12,"temp_min":5,"humidity":500,"condition":"Rain"}
```

DELETE Delete forecast

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Deletes the forecast saved in . Expects 204 No Content.

AUTHORIZATION Bearer Token

Token

POST Add forecast as normal user

{{base_url}}/locations/{{city_id}}/forecasts

Attempts to add a forecast as a normal (non-admin) user. Expects a 403 Forbidden response.

HEADERS

Content-Type	application/json

Body raw (json)

json


```
{"date": "", "temp_max": 12, "temp_min": 5, "humidity": 80, "wind_kph": 20, "precip_mm": 3,
```


Observations

Endpoints for submitting and deleting observed weather data for a location.

POST Add observation

{{base_url}}/locations/{{city_id}}/observations

Submits a new weather observation for the tracked city. Saves the new observation ID to .

AUTHORIZATION Bearer Token

Token

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user bob

Body raw (json)

json

```
{"condition": "Clear", "date": "", "temp_max": 25, "temp_min": 18}
```

DELETE Delete observation

{{base_url}}/locations/{{city_id}}/observations/

Deletes the observation saved in . Expects 204 No Content.

AUTHORIZATION Bearer Token

Token

DELETE Delete someone else's observation

{{base_url}}/locations/{{city_id}}/observations/

Attempts to delete another user's observation using a different user's token. Expects a 403 Forbidden response.

Analytics

Endpoints for querying aggregated weather statistics and rankings across locations.

GET Location stats

{{base_url}}/locations/{{city_id}}/stats

Returns aggregated weather statistics for the tracked city.

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{{base_url}}/locations/{{city_id}}/accuracy

Returns the forecast accuracy score for the tracked city.

GET Hottest cities

{{base_url}}/analytics/hottest?limit=5

Returns the top cities ranked by maximum temperature.

PARAMS

limit	5

GET Rainiest cities

{{base_url}}/analytics/rainiest?limit=5

Returns the top cities ranked by total precipitation.

PARAMS

limit	5

GET Windiest cities

{{base_url}}/analytics/windiest?limit=5

Returns the top cities ranked by average wind speed.

PARAMS

limit	5

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{{base_url}}/analytics/hottest?country=UK

Returns the hottest cities filtered to the UK.

PARAMS

country	UK

GET Condition breakdown

{{base_url}}/analytics/condition-breakdown

Returns a breakdown of weather conditions (e.g. Rain, Clear) across all observations.

GET Stats invalid id

{{base_url}}/locations/abc/stats

Requests stats for a non-numeric location ID. Expects a 400 or 404 error response.

Tracker

Endpoints for tracking and untracking cities, and viewing the user's tracker dashboard.

POST Track a city

{{base_url}}/users/me/tracked/{{city_id}}

Adds the city ({{city_id}}) to the authenticated user's tracked cities list.

GET My tracker dashboard

{{base_url}}/users/me/tracker

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POST Track without login

{{base_url}}/users/me/tracked/{{city_id}}

Attempts to track a city without an Authorization header. Expects 401 Unauthorized.

POST Track invalid id

{{base_url}}/users/me/tracked/abc

Attempts to track a city using a non-numeric ID. Expects a 400 or 404 error response.

DELETE Untrack a city

{{base_url}}/users/me/tracked/{{city_id}}

Removes the city ({{city_id}}) from the authenticated user's tracked cities list.

DELETE Untrack again

{{base_url}}/users/me/tracked/{{city_id}}

Untracks the city a second time to verify idempotent behaviour.

Alerts

Endpoints for creating, editing, listing, and deleting weather threshold alerts.

POST Create alert

{{base_url}}/alerts

Generated from cURL: curl -X POST {{base_url}}/alerts -H "Content-Type: application/json" -d '{"location_id": "{{city_id}}", "metric": "temp_max", "operator": "gt", "threshold": 5}'

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Content-Type application/json

Body raw (json)

```
json

{"location_id": "{{city_id}}", "metric": "temp_max", "operator": "gt", "threshold": 5}
```

POST Create alert bad metric

{{base_url}}/alerts

Generated from cURL: curl -X POST {{base_url}}/alerts -H "Content-Type: application/json" -d '{"location_id": "{{city_id}}", "metric": "sunshine", "operator": "gt", "threshold": 5}'

HEADERS

Content-Type application/json

Body raw (json)

```
json

{"location_id": "{{city_id}}", "metric": "sunshine", "operator": "gt", "threshold": 5}
```

GET List my alerts

{{base_url}}/alerts

Generated from cURL: curl -X GET {{base_url}}/alerts

GET Triggered alerts

{{base_url}}/alerts/triggered

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PUT Edit alert

{{base_url}}/alerts/

Generated from cURL: curl -X PUT {{base_url}}/alerts/ -H "Content-Type: application/json" -d '{"threshold": 20}'

HEADERS

Content-Type	application/json

Body raw (json)

```json
{"threshold": 20}
```

DELETE Delete alert

{{base_url}}/alerts/

Generated from cURL: curl -X DELETE {{base_url}}/alerts/

DELETE Delete alert again

{{base_url}}/alerts/

Generated from cURL: curl -X DELETE {{base_url}}/alerts/